

# DEFENDED DYNAMIC STOPPING ENVELOPE: REACTION LAG VS. KINETIC BRAKING

Linear Sensitivity to Computational Latency vs. Quadratic Sensitivity to Velocity:  $d_{stop}(v_0, \Delta t) = v_0 \Delta t + v_0^2 / (2a) + \delta_{overhead} \leq D_{clear}$

## 1. LINEAR REACTION TRAVEL

$$d_{react} = v_0 \cdot \Delta t_{delay}$$

Unguided coasting during compute latency

## 2. QUADRATIC BRAKING DISTANCE

$$d_{brake} = v_0^2 / (2 \cdot a_{brake})$$

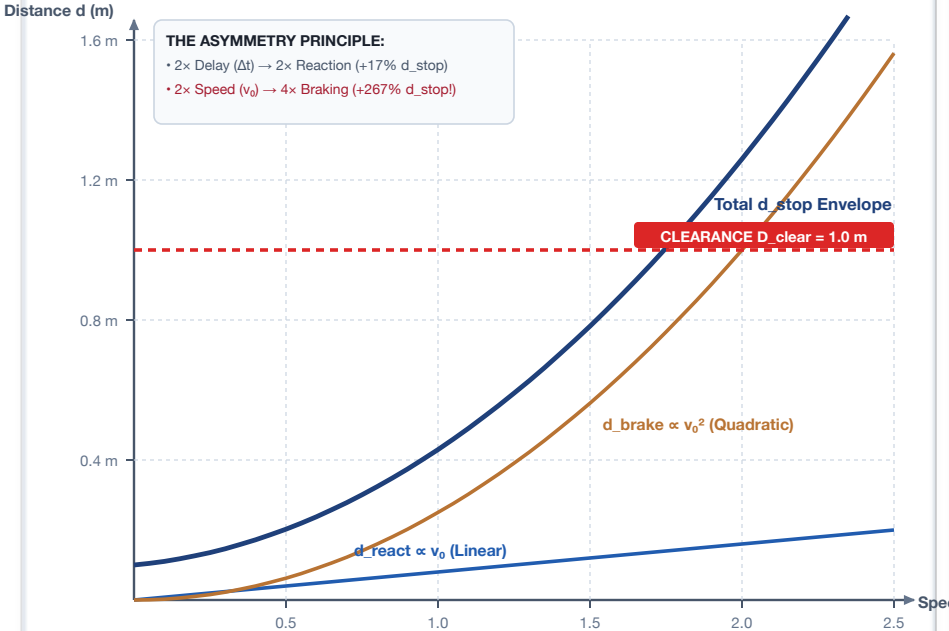
Work-energy dissipation:  $E_k = \frac{1}{2} m v_0^2$

## 3. MAX PERMITTED VELOCITY

$$v_{max} = -a \Delta t + \sqrt{(a \Delta t)^2 + 2aD_{avail}}$$

Defended clearance bound:  $d_{stop} \leq D_{clear}$

### ASYMMETRIC PARAMETER SENSITIVITY CURVES



### PHYSICAL SCENARIO TRAJECTORIES ( $D_{clear} = 60.0$ cm)

#### CASE 1: Nominal Speed & Latency ( $v_0 = 1.0$ m/s, $\Delta t = 50$ ms)

5cm

Braking 25.0 cm

Margin +20.0 cm ✓

Total Stop  $d_{stop} = 40.0$  cm  $\leq 60.0$  cm (CERTIFIED SAFE)

#### CASE 2: Tail Latency Spike ( $v_0 = 1.0$ m/s, $\Delta t = 200$ ms)

Reaction 20.0 cm (+15cm unguided!)

Braking 25cm

+5cm

Total Stop  $d_{stop} = 55.0$  cm (CLEARANCE MARGIN COMPRESSED TO 5 cm)

#### CASE 3: Speed Increase + Tail Latency ( $v_0 = 1.5$ m/s, $\Delta t = 200$ ms)

Reaction 30.0 cm

Braking 30cm

Total Stop  $d_{stop} = 86.3$  cm  $> 60.0$  cm (CATASTROPHIC IMPACT BREACH!)

### FIRST-PRINCIPLES PHYSICAL AI DESIGN LAW

Software models propose speed setpoints; the Body enforces quadratic kinetic momentum. Safe operation requires budgeting P99.9 latency and defending physical clearance.